



Learning objectives

By the end of this project, you will be able to:

- Load and profile real-world data read a messy dataset with the correct encoding, then use head, shape, info, describe, and value_counts to assess its size, structure, and quality.
- Apply the full pandas wrangling toolkit demonstrate all 21 data-cleaning techniques (creating, selecting, filtering, sorting, replacing, renaming, handling missing values, dropping rows/columns/duplicates, grouping, aggregating, applying functions, resampling, concatenating, and merging) on a single dataset.
- Diagnose and resolve common data problems identify and handle missing values, duplicate rows, inconsistent categories, invalid prices/quantities, and cancellations/returns.
- Make and justify cleaning decisions, choose defensible strategies (what to drop, keep, or repair), keep raw and cleaned data separate, and treat dataframes as immutable objects.
- Turn clean data into insight engineer features and aggregate the data to answer real business questions about sales, products, markets, and customers.
- Communicate to a non-technical audience present finding clearly with short interpretations and at least two supporting charts.
- Produce reproducible, version-controlled work document your process and manage your code and outputs in a GitHub repository that another person can clone and run end to end.

The scenario

You have just been hired as a junior data science team at **HealthGuard Insurance**, a mid-sized health insurance company. The company currently prices its insurance plans using outdated actuarial tables, and management believes this is causing two problems:

1. **Underpricing** high-risk customers (the company loses money on them).
2. **Overpricing** low-risk customers (they leave for cheaper competitors).

The **Head of Pricing (your stakeholder, Ms. Sarah Nabil)** has asked your team to build a data-driven solution using the company's historical customer records. The dataset contains each customer's age, sex, BMI, number of children, smoking status, region, and the actual medical charges billed to them. Link to the dataset <https://www.kaggle.com/datasets/mirichoi0218/insurance>

2. Stakeholder Requirements

Ms. Nabil has given you the following requirements, written in her own (non-technical) words. Your job is to translate them into machine learning tasks.

Requirement 1 "Understand our customers first" (EDA)

Before you build anything, show me what drives medical costs. Which customer attributes matter most? Are there any surprising patterns? I want visuals I can show to the board."

Deliverable: An exploratory data analysis with at least 5 meaningful visualizations and a short-written summary of insights (effect of smoking, BMI, age on charges). Handle any data cleaning/encoding needed.

Requirement 2 "Predict the medical charges" (Regression)

I need a tool that estimates how much a new customer will cost us per year, so we can price their premium fairly.

Deliverable: Build, compare, and evaluate the following regression models to predict charges:

#	Model	Notes
1	Simple Linear Regression	Use the single best predictor you found in EDA
2	Multiple Linear Regression	All features
3	Polynomial Regression	Try degrees 2–4; discuss overfitting
4	Ridge Regression (L2)	Tune alpha
5	Lasso Regression (L1)	Tune alpha; report which features it eliminated
6	Support Vector Regression (SVR)	Try at least two kernels; remember to scale features
7	Decision Tree Regression	Control max_depth; visualize the tree if possible

Evaluation: Use a proper train/test split. Compare all models with **MAE, MSE, RMSE, and R^2** in a single summary table, and recommend ONE model to the stakeholder with justification.

Requirement 3 "Flag the expensive customers" (Classification)

Honestly, I don't always need the exact dollar amount. Sometimes I just need to know: is this customer going to be *expensive* or not? Our finance team defines 'expensive' as anyone whose annual charges exceed the media.

Deliverables

- Your work is delivered in two places: a **GitHub repository** (all the technical work) and a **zipped submission to Canvas**.

GitHub repository must contain:

- The cleaning notebook (.ipynb)
- A technical report documenting your full workflow: data cleaning decisions and rationale, exploratory analysis with key insights, model development and evaluation, and a final recommendation to the stakeholder.
- A README.md explaining what the project does, how to download the dataset from Kaggle, how to run the notebook, and a brief summary of your findings.
- Requirements.txt listing the Python libraries (and versions) needed to reproduce your work.

Submit to Canvas:

- The **link to your GitHub repository**, and
- A **single zipped folder** of the repository.